
2.1 INTRODUCTION

In this Unit we will learn about the Keynesian model that was first developed by the British economist John Maynard Keynes in his famous book ‘The General Theory of Employment, Interest and Money’, published in 1936. This model differs in several respects from the classical approach to macroeconomics that you studied in the previous Unit. The Keynesian model allows for the existence of unemployment as well as rigidity in wages and prices. It recognizes that lack of effective demand may prevent the attainment of full employment of resources. Also, it emphasizes the need for government intervention – as the market mechanism would not be able to restore full employment equilibrium due to rigidities in wage rate and prices.

2.2 COMPONENTS OF AGGREGATE DEMAND

We observe that Keynesian economics gained importance against the backdrop of the Great Depression of the 1930s, when developed economies like the USA and Great Britain faced a deep and prolonged recession, with rising unemployment, falling revenues and profits of firms and rise in unutilized capacity. In this situation the market mechanism failed to restore equilibrium and the Keynesian policy of raising aggregate demand via increase in public expenditure played a crucial role in addressing the problem.

Aggregate demand is the planned expenditure on goods and services by all the economic agents, such as households, firms and government. There are three main components of aggregate demand in a closed economy: (i) *planned* consumption expenditure (C) of households, (ii) *planned* investment expenditure (I) of firms, and (iii) expenditure incurred by the government on goods and services (G). Throughout our discussion in this Unit, each of C, I and G are assumed to be in real terms (i.e., no change in prices).

2.2.1 Consumption Function

Keynes proposed the idea of a Consumption function which gives the relation between *planned* consumption expenditure of households (C) and aggregate income (Y). It can be represented by the following linear equation:

$$C = a + bY \quad \dots (2.1a)$$

In the presence of the government, a part of income would be paid in taxes (T). In that case the consumption function would be represented as a function of disposable incomes ($Y_d = Y - T$), that is,

$$C = a + b(Y - T) = a + bY_d \quad \dots (2.1b)$$

For the discussion below, for simplicity, we assume that there is no government, so that $T = 0$ and $Y = Y_d$.

The slope of the consumption function ‘ b ’ is assumed to be a positive fraction (i.e., $0 < b < 1$). It means that the consumption function can be represented as an

upward sloping straight line as in Fig. 2.1. It indicates that planned consumption expenditure increases with an increase in the level of aggregate income.

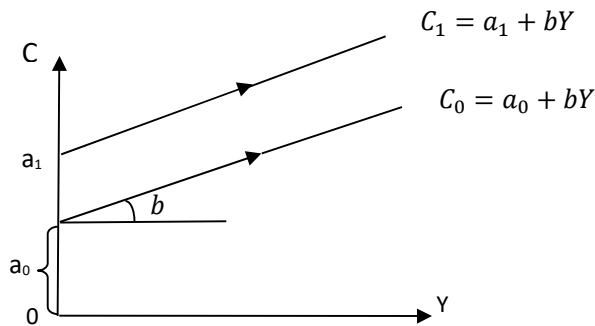


Fig. 2.1: Consumption Function

Note that ‘ b ’ is defined as the *marginal propensity to consume* (MPC) and it represents the change in consumption induced by a unit change in income. The MPC is assumed to be less than one. If there is an increment in income by ΔY , the induced increase in consumption ΔC is a fraction of ΔY (i.e., $\Delta C = b\Delta Y$ and $\Delta C/\Delta Y = b < 1$). It means that the entire increment in income is **not** spent on consumption, a fraction $(1 - b)$ is saved. It means total saving out of the incremental income is $(1 - b)\Delta Y$.

Let us consider the case where the consumption function is given by $C = 20 + 0.25Y$. If aggregate income increases by Rs.100 (i.e., $\Delta Y = 100$), then Rs. 25 ($\Delta C = 0.25 \times 100$) is the additional consumption and Rs.75 $[(1 - 0.25) \times 100]$ is additional saving of households out of the incremental income. Similarly, if income declines by Rs.100, then consumption would decline by Rs. 25 only, while saving would decline by Rs. 75.

The intercept of the consumption function ‘ a ’ represents autonomous consumption, with $a > 0$. From equation (2.1a), you can see that when $Y = 0$, we have $C = a$. It implies that consumption occurs even when income is zero. This consumption is financed by drawing down past saving or by ‘dis-saving’. An increase in ‘ a ’ represents increase in planned consumption of households at each level of income; this can happen, for instance, when consumer confidence increases (i.e., households’ expectations about the future turn buoyant). You can see in Fig. 2.1 that an increase in the intercept (from a_0 to a_1) results in a parallel, upward shift of the consumption function, from C_0 to C_1 .

Aggregate consumption has two components: (i) an autonomous component ‘ a ’ that does not depend on income, and (ii) an ‘induced’ component ‘ bY ’ that depends on aggregate income levels. An increase (decrease) in Y induces an increase (decrease) in planned consumption expenditure by households.

The ratio of aggregate consumption expenditure to aggregate income $\left(\frac{C}{Y}\right)$ is defined as the *average propensity to consume* (APC). From equation (2.1a) you

can see that $APC = C/Y = a/Y + b$. Note that APC will decline as Y increases, while MPC remains constant.

2.2.2 Investment Function

The investment function (I) represents *planned* investment expenditure undertaken by firms on machinery, factories and other such items that add to productive capacity of the economy. In the simplest version of the Keynesian model investment is assumed to be ‘autonomous’, i.e., independent of the level of aggregate income (Y). In this case the investment function is represented by the equation as below.

$$I = \bar{I} \quad \dots (2.2a)$$

In equation (2.2a) a ‘bar’ over the variable investment ($\bar{I}$) variable indicates a fixed amount. However, we can easily relax this assumption and allow investment to be a function of aggregate output. In this case the investment function is:

$$I = c + dY \quad \dots (2.2b)$$

where $c > 0$ and $d > 0$; ‘ c ’ is the autonomous part of investment, d is the *marginal propensity to invest* (MPI) and ‘ dY ’ represents investment that is induced by aggregate income.

Fig. 2a below depicts an autonomous investment function, while Fig. 2.2b depicts an induced investment function which has a component that is induced by income.



Fig. 2.2a: Autonomous Investment

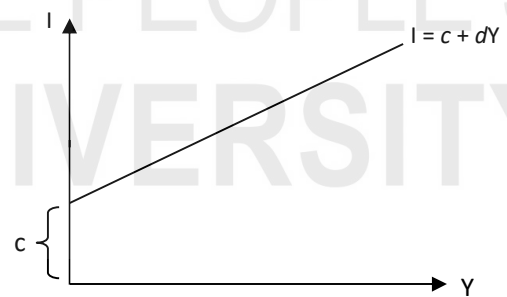


Fig. 2.2b: Induced Investment

Note that the term investment refers to planned expenditure by firms on new production equipment (e.g., machinery, building, etc.) that add to productive capacity. It does *not* refer to buying of corporate bonds or shares in a company or opening a fixed deposit account in a bank.

An important variable that affects firms’ decision to invest is the rate of interest. As the rate of interest increases, the cost of borrowing goes up, resulting in a reduction in the firms’ planned investment expenditure. The inverse relation between investment and interest rate can be further explained as follows: